Whether Mean Token Savings Hide Errors in the Long Tail

Working Title

Whether Mean Token Savings Hide Errors in the Long Tail.

Problem Statement

SketchThinker-R1's token count and its efficiency ratio are means, and reasoning length is heavy-tailed. Sprint or Delve (IJCAI/ECAI 2026) already fits length with a log-normal and softens the penalty on long traces; it does not measure the quantile-conditional accuracy of a style reward on these multimodal benchmarks.

Proposed Solution

This plan does not change the training penalty. Sketch-style training and the paper's vanilla R1 baseline are reproduced on Qwen2.5-VL-7B with the paper's data. Each benchmark is split into quintiles of the baseline's reasoning length. Accuracy gap, token ratio, and end-to-end latency are reported per quintile, with multiple seeds and confidence intervals. The primary contrast is items above the baseline's 90th percentile against items below the median. The mean reduction is reported only as a reference. The proposition fails if the accuracy gap does not grow across quintiles. A loss in the tail is not, by itself, evidence that the judge should see the image.

Key Research Questions

- Does the accuracy gap grow from the shortest quintile to items above the 90th percentile?
- Does the token reduction in the upper quintile stay as large as the mean reduction?
- Are upper-quintile losses larger than the binomial uncertainty of that slice?

Expected Outcome

The study will give a quantile-specific reading of the reported mean reduction, including the slices where that mean misleads.

Comparison

Summary	What the paper uses	This plan
Length	Mean reasoning tokens	Quintiles of the baseline length
Accuracy	One mean per benchmark	A gap inside each quintile
Result that refutes it	—	The gap does not grow across quintiles